AHSAN BASHARAT

Electrical Engineer | Embedded Systems & Electronics Engineer

ahsanbasharatali38@gmail.com | +92 331 4249022 | linkedin.com/in/ahsan-basharat-35911129b

PROFESSIONAL SUMMARY

Electrical Engineering student with hands-on experience in embedded systems, firmware development, electronics, industrial testing, and quality control. Proficient in C/C++, object-oriented programming, STM32, Arduino, circuit design, MATLAB, PCB design, hardware interfacing, and communication systems, gained through internships in embedded engineering, transformer inspection, and energy meter production. Seeking to contribute as an Embedded, Firmware, Electronics, or Electrical Engineer on impactful engineering projects.

EDUCATION

BS Electrical Engineering *2023 – 2027*

Information Technology University (ITU), Lahore, Pakistan

PROFESSIONAL EXPERIENCE

Intern, Inspection & Quality Control – Distribution Transformers *Jul 2025 – Aug 2025*

Pak Elektron Limited (PEL), Lahore, Pakistan

- Studied the end-to-end distribution transformer manufacturing process, including design, winding, core stacking, assembly, and testing.
- Observed transformer oil testing per IEC 60422/IEC 60296, including breakdown voltage, moisture content, flash point, and dissolved gas analysis (DGA).
- Assisted in final routine electrical testing, including winding resistance, no-load loss, short-circuit impedance, and dielectric withstand tests.

Intern, Embedded Engineering *Jul 2026 – Sep 2026*

Fooran Technology

- Developed embedded applications using C/C++ and object-oriented programming (OOP) principles.
- Gained hands-on experience with STM32 Blue Pill and STM32F103RB microcontrollers.
- Worked with GPIO, UART, ADC, timers, and PWM for hardware interfacing and firmware development.

Intern, Energy Meter Production (EMP) *Aug 2026 – Sep 2026*

Pak Elektron Limited (PEL), Lahore, Pakistan

- Gained hands-on experience with single-phase and three-phase energy meters, including assembly, inspection, and testing.
- Worked with meter communication systems, including IEC 62056 (DLMS/COSEM), for data communication and interfacing.
- Performed functional and performance testing per IEC 62052 and IEC 62053 standards to verify meter accuracy and reliability.

TECHNICAL SKILLS

Programming & Software: C, C++, Python, MATLAB, STM32CubeIDE, STM32CubeMX, GNU Radio, LabVIEW,Altium Design

Embedded Systems & Hardware: STM32, Arduino, UART, GPIO, ADC, Timers, PWM, Sensor & Actuator Interfacing, Circuit Analysis, PCB Prototyping, USRP

Design & Simulation: PCB Design, Altium Designer, MATLAB/Simulink, Proteus, Multisim, LTspice

Engineering Areas: Embedded Systems, Firmware Development, Electronics, Circuit Design, Hardware Interfacing, Communication Systems, Image Processing

Languages: English – Intermediate | Urdu – Fluent

PROJECTS

Industrial Vibration Monitoring & Self-Calibration System: Designed an electronics-based system for motor vibration detection, signal conditioning, and self-calibration.

Transformer Design Project: Designed and analyzed a transformer, covering electrical specifications, winding parameters, and core design.

Electronics & Embedded Engineering Projects: Completed practical projects in circuit design, STM32 microcontrollers, sensors, actuators, and hardware-software integration.

University Management System (C++): Developed a C++ system for managing student and academic records using object-oriented programming.

AI-Based Traffic Challan Processing: Built a machine-learning system for automated traffic challan processing using image processing and computer vision.

Controlled GPS/SDR Framework (GNU Radio & USRP): Developed an SDR-based communication framework for studying wireless communication systems.

Home Automation System (STM32): Designed a home automation system to control lights, fans, and appliances using STM32 and sensors.

STM32 DC-DC Converter: Designed and implemented step-up and step-down DC-DC converters using an STM32 microcontroller.

ACHIEVEMENTS

Automated Battery Health & Performance Testing System: Project accepted for display at the Pak Elektron Limited (PEL) Project Exhibition.